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Placenta NMR Imaging

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Abstract

The **purpose** of this work was to test whether the results of **diffusion** and **perfusion** obtained in the work of Maiuro et al. [1] depended on the static magnetic field strength ($1.5T$ vs $3.0T$).

Precisely, we investigate the diffusion and perfusion properties of the **fetal and maternal placenta** using diffusion-weighted imaging (DWI). We analyzed 15 placentas with a $1.5T$ *Siemens Avanto* clinical scanner and 7 placentas with a $3T$ *Siemens Prisma* clinical scanner. The two-compartment perfusion, the IVIM, and the monoexponential models have been employed for estimating the various parameters.

Moreover, the **results** indicate distinct diffusion and perfusion characteristics between the fetal and maternal regions of the placenta, with the fetal part displaying higher values for both the total perfusion fraction f and the slowest perfusion fraction f_2 , reflecting increased metabolic activity and nutrient exchange necessary for fetal development. Conversely, the maternal part showed lower values, consistent with its role in supplying blood to the placenta.

The Signal-to-Noise Ratio (**SNR**) analysis revealed that the $3T$ acquisitions provide significantly higher SNR compared to the $1.5T$, enhancing the sensitivity and contrast of the imaging data.

The detailed data analysis and model fitting were performed using custom *Python* codes available at [1.5T code](#) and [3T code](#).

1 Introduction

1.1 The Physical Basis of NMR

Nuclear Magnetic Resonance (NMR) is a highly effective analytical method used to elucidate the structure, behavior, and chemical context of molecules. This technique leverages the magnetic characteristics of specific atomic nuclei. Essentially, NMR operates on the principle of the interaction between an external magnetic field and the intrinsic nuclear spin of many nuclei [2].

Nuclear Spin and Magnetic Moments

Atoms consist of a central nucleus surrounded by electrons. The nucleus itself contains protons and neutrons. Some nuclei display an important property known as **spin**, making them behave like tiny magnets. Moreover, spin is a quantum mechanical property, and nuclei with a nonzero spin have a corresponding magnetic moment. Specifically, nuclei with an odd number of protons or neutrons possess a net spin¹ and, thus, a magnetic moment μ .

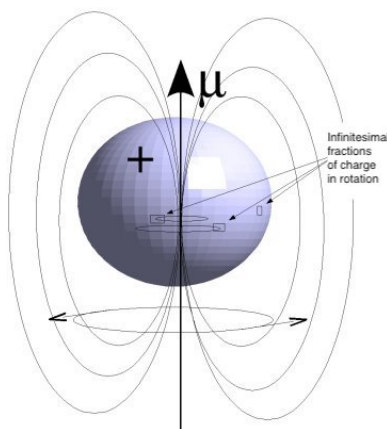


Figure 1: Illustration of nuclear spin and magnetic moment. Fractions of charge in rotation create a magnetic moment μ [3].

When subjected to an external magnetic field $\vec{B}_0$, these nuclear magnets align either along or opposite to the field direction. This alignment is *quantized*, meaning the magnetic moment can only adopt specific orientations relative to the external field. For example, a nucleus with a spin $s = \frac{1}{2}$, like hydrogen-1 (^1H), has two possible orientations: parallel (lower energy) and antiparallel (higher energy) to $\vec{B}_0$.

¹Net spin is the overall spin of a nucleus, resulting from the combined spins of its protons and neutrons.

Energy Levels and Resonance

When an external magnetic field $\vec{B}_0$ is applied, the magnetic moments of nuclei undergo a torque that causes them to **precess** around the direction of the field at a specific frequency, known as the Larmor frequency, ω_0 . This frequency is determined by the equation [2]:

$$\omega_0 = \gamma B_0 \quad (1)$$

where γ represents the gyromagnetic ratio, a nucleus-specific constant.

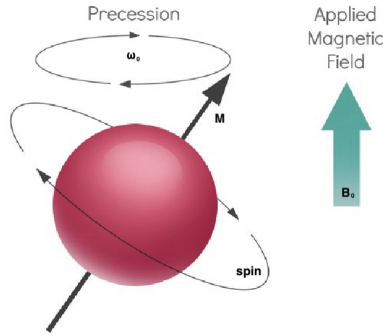


Figure 2: The alignment of nuclei with an applied magnetic field is not stagnant: nuclei precess around the direction of B_0 [4].

The energy difference ΔE between the parallel and antiparallel states is directly proportional to the magnetic field strength and can be expressed as [2]:

$$\Delta E = \hbar \omega_0 = \hbar \gamma B_0 \quad (2)$$

with $\hbar$ denoting the reduced Planck constant.

This energy difference matches the energy of a photon capable of inducing **transitions between these states**.

NMR spectroscopy relies on detecting these transitions. When a **radio frequency (RF) pulse** is applied at the Larmor frequency, it perturbs the nuclear spins from their alignment with the main magnetic field B_0 . This RF pulse causes the spins to oscillate between “up” and “down” states, a phenomenon known as *Rabi oscillations*. The absorption of energy during these transitions is known as **resonance**, and it forms the basis of the NMR signal. Essentially, the RF pulse induces transitions between energy levels by flipping the spins back and forth, and the resonance detected during this process is crucial for generating the NMR signal [4].

Relevant timing parameters in NMR

Once nuclei are excited to a higher energy state, they gradually return to equilibrium, releasing energy in the process. This return to equilibrium is controlled by two key

relaxation mechanisms: longitudinal (spin-lattice) relaxation and transverse (spin-spin) relaxation.

Longitudinal Relaxation (T_1)

Longitudinal relaxation involves the transfer of energy from nuclear spins to the surrounding lattice (environment). T_1 is the time constant for the physical processes responsible for the relaxation of the components of the nuclear spin magnetization vector $\vec{M}$ parallel to the external magnetic field, $\vec{B}_0$ (which is conventionally designated as the z -axis). Therefore, in general [4]:

$$M_z(t) = M_{z,\text{eq}} - [M_{z,\text{eq}} - M_z(0)] e^{-t/T_1} \quad (3)$$

If the magnetization vector $\vec{M}$ is rotated into the xy -plane, the initial longitudinal component is $M_z(0) = 0$. Then, the return to equilibrium can be expressed by:

$$M_z(t) = M_{z,\text{eq}} \left(1 - e^{-t/T_1}\right) \quad (4)$$

where $M_{z,\text{eq}}$ denotes the equilibrium magnetization along the z -axis, and T_1 is the time constant for spin-lattice relaxation.

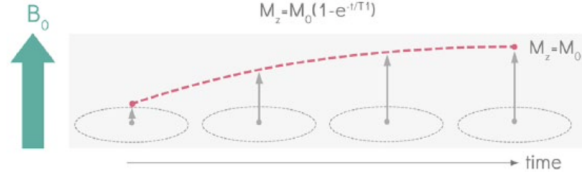


Figure 3: When a RF pulse is applied, the magnetization vector, initially aligned with $\vec{B}_0$, is tipped to the transverse plane (perpendicular to $\vec{B}_0$). In this state, the longitudinal component $M_z = 0$. After the pulse is turned off, the magnetization vector gradually returns to its original alignment. T_1 is therefore the time constant for this regrowth of the longitudinal component M_z back to its original alignment, known as thermal equilibrium (at equilibrium, the magnetization amplitude is $M_0 = M_{z,\text{eq}}$) [4].

Understanding this process is crucial for determining how quickly spins can realign with the external magnetic field after excitation.

Transverse Relaxation (T_2)

The transverse relaxation time, T_2 , quantifies the rate at which the transverse magnetization (M_{xy}), perpendicular to the static magnetic field $\vec{B}_0$, decays due to spin-spin interactions and magnetic field inhomogeneities. For instance, the transverse magnetization that starts at $M_{xy}(0)$ at time zero diminishes to zero as it returns to equilibrium, following the relationship [4]:

$$M_{xy}(t) = M_{xy}(0)e^{-t/T_2} \quad (5)$$

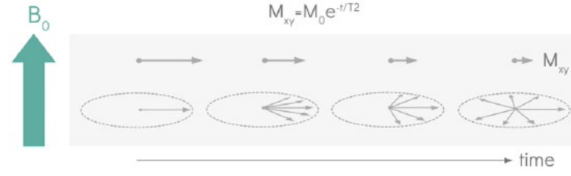


Figure 4: The transverse magnetization (M_{xy}) of protons, which is perpendicular to $\vec{B}_0$, precesses in phase around this field. Small variations in $\vec{B}_0$ cause different precession rates, leading to a loss of coherence. This loss of coherence causes M_{xy} to decay [4].

Echo Time (TE) and Repetition Time (TR)

Echo Time (TE) is the interval between the delivery of the radiofrequency (RF) pulse and the peak of the echo signal in MRI. TE is a critical parameter that significantly influences the contrast and quality of the images. It determines the amount of transverse relaxation that occurs before the signal is measured. Longer TE values permit more time for T_2 decay, thereby enhancing the differences in T_2 properties between tissues and providing greater contrast in T_2 -weighted imaging.

Repetition Time (TR) is the interval between successive RF pulses. TR influences the degree of longitudinal relaxation that occurs between pulses. Short TR values can reduce the time available for T_1 recovery, leading to differences in signal intensity based on T_1 properties, thereby enhancing T_1 -weighted imaging contrast. TR is a key factor in determining the overall scan time and affects the signal-to-noise ratio and image resolution.

Spin Echo

Spin Echo is a fundamental MRI sequence used to refocus spins that have dephased due to inhomogeneities in the magnetic field. After an initial 90 degree RF pulse, which tips the net magnetization into the transverse plane, a 180-degree RF pulse is applied at the time equal to half the echo time. This 180 degree pulse inverts the spins, causing them to refocus and form an echo at the time TE. The spin echo sequence is crucial for reducing artifacts and improving image quality by correcting for inhomogeneities and allowing accurate measurement of T_2 relaxation times.

Diffusion Signal Acquisition

Diffusion-weighted imaging (DWI) measures the random Brownian motion of water molecules within tissues. The diffusion signal is acquired using strong gradient pulses in addition to the spin echo sequence. The initial gradient pulse dephases the spins, while the subsequent gradient pulse rephases them. The degree of rephasing depends on the displacement of the water molecules during the time between the gradient pulses, providing information about tissue microstructure.

In MRI, the Fourier transform is used to convert raw data from the time domain to the frequency domain, generating the final image. The motion of water molecules can be described using various models. In free water, the monoexponential decay model proposed by Stejskal and Tanner is appropriate (see eq. 7). However, in complex tissues like the placenta, more sophisticated models are needed to account for the heterogeneous environment. In this study, we evaluate two additional models to the one proposed by Stejskal and Tanner: the two-compartment perfusion model (eq. 6) and the intravoxel incoherent motion (IVIM) model (eq. 8), to better understand diffusion and perfusion properties in the placenta.

1.2 Basic Information on Human Placenta

The placenta plays a crucial role in transferring oxygen and nutrients from the mother to the fetus during pregnancy. The human placenta is of the hemochorial type, meaning that maternal and fetal blood do not mix. It consists of two distinct parts: the chorionic plate and the basal plate.

The chorionic plate, the **fetal portion**, contains villus trees (whose surface is covered by trophoblastic cells²) filled with fetal blood. These villi facilitate the exchange of nutrients, oxygen, and waste products with the maternal blood. On the **maternal side**, the basal plate encases the intervillous space where maternal blood flows. This arrangement allows for efficient transfer of essential substances from the mother to the fetus and removal of waste products from the fetal blood to the maternal circulation.

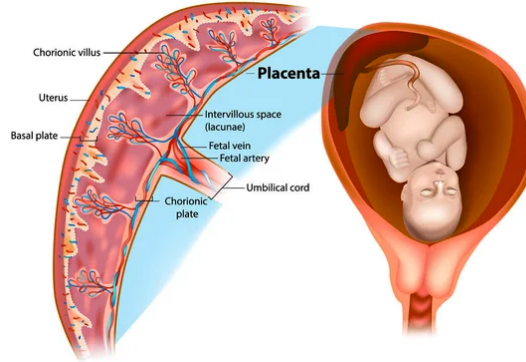


Figure 5: Placenta schematization [4].

This highly specialized structure ensures the fetus receives the necessary resources for growth and development while keeping maternal and fetal blood separate. The placenta also produces hormones that support pregnancy and fetal growth, such as human chorionic gonadotropin (hCG), progesterone, and estrogen. Additionally, it

²Trophoblasts are the cells constituting fetal components in the placenta and fetal membrane. They serve multiple functions to maintain the mammalian pregnancy, such as the supply of nutrients and gases, secretion of pregnancy-associated hormones, and protection of the semi-allogeneic fetus from immune rejection [5].

acts as a barrier, protecting the fetus from potential infections and harmful substances present in the maternal blood.

1.3 Models

The physiology of the placenta involves at least two primary **perfusion**³ compartments. One compartment involves the exchange of substances through the trophoblastic cells between the maternal and fetal sides, while the other pertains to the blood perfusion within the villus trees. It is plausible that the perfusion rates of these two compartments differ by at least an order of magnitude. Additionally, the primary source of **diffusion** in the placenta is the concentration gradients of gases (such as O₂ or CO₂), nutrients, and waste products between the maternal blood in the intervillous space and the fetal blood in the chorionic villi. This diffusion occurs across the placental barrier, allowing for the exchange of essential substances to support fetal development.

The model proposed by Maiuro et al. [1], namely the **two-compartment perfusion**, is the following:

$$S(b) = S(0) \left[f_1 e^{-(D_1^* + D_2^* + D)b} + f_2 e^{-(D_2^* + D)b} + (1 - f_1 - f_2) e^{-Db} \right] \quad (6)$$

where D is the pseudo-diffusion coefficient relating to the maternal blood flowing inside the intra-villous space, D_1^* is the fastest pseudo-perfusion coefficient given by the blood flowing inside micro-vessels and villus trees, D_2^* is the slower pseudo-perfusion coefficient given by the exchange of nutrients between the maternal and the fetal compartments, f_1 is the fastest perfusion fraction and f_2 is the slowest perfusion fraction related to the trophoblastic cells. The choice to add D_2^* in the fast perfusion compartment was because it could not be distinguished from the fastest D_1^* relating to the villi's vasculature, thus D_2^* should contribute to the actual fast perfusion compartment. We will assume this model concerning the perfusion in “our” placentas.

In order to reduce the number of independent parameters, the diffusion coefficient D underwent initial estimation by employing a **monoexponential** model [2] on diffusion-weighted imaging (DWI) data at high b -values (where $b \geq 200$ s/mm²):

$$S(b) = S(0) e^{-bD} \quad (7)$$

Subsequently, given the placenta's distinct dual perfusion processes contributing to overall perfusion dynamics, the **intravoxel incoherent motion (IVIM)** model [6] was utilized to derive the total perfusion fraction $f = f_{\text{IVIM}} = f_1 + f_2$, by fitting the equation (with D “fixed”):

$$S(b) = S(0) \left[f e^{-bD^*} + (1 - f) e^{-bD} \right] \quad (8)$$

³Perfusion - in placenta - is the process by which maternal blood flows through the placental blood vessels, delivering oxygen and nutrients to the fetus and removing waste products.

The two-compartment perfusion model was utilized after estimating D and f from the preceding models, namely the monoexponential and IVIM models.

Further Parameters Explanation

The **Diffusion Coefficient** (D) is derived from the monoexponential model (eq. 7) and it is associated with the maternal blood flow within the intra-villus space. It characterizes the overall diffusion process occurring within the placenta. The role of D is to provide an estimate of how substances diffuse through the placental tissue.

The **Total Perfusion Fraction** (f) is derived from the IVIM model (eq. 8) and represents the combined contribution of different perfusion compartments within the placenta.

The **Slowest Perfusion Fraction** (f_2) is derived from the two-perfusion model (eq. 6) and represents the fraction of perfusion related to the slowest compartment, which corresponds to the exchange of nutrients between the maternal and fetal compartments through the trophoblastic cells, as already mentioned before. This parameter characterizes the slowest exchange process occurring in the placental tissue.

The **Fastest Perfusion Fraction** (f_1 , derived from eq. 6) represents the fraction of perfusion related to the fastest compartment, which corresponds to the blood flowing inside micro-vessels and villus trees. This parameter characterizes the rapid exchange processes occurring within the placental vasculature.

On the **fetal side of the placenta**, the perfusion fraction (f) is expected to be higher due to the increased metabolic activity and nutrient exchange necessary for fetal development. Similarly, the slowest perfusion fraction (f_2) is anticipated to be higher, as the trophoblastic cells are actively involved in nutrient exchange between the maternal and fetal compartments.

In contrast, on the **maternal side of the placenta**, the perfusion fraction (f) is expected to be lower compared to the fetal side, as its primary role is to supply blood to the placenta rather than to engage in direct nutrient exchange. The slowest perfusion fraction (f_2) is expected to be reduced on the maternal side, indicating a diminished role in the direct nutrient exchange process.

2 Materials and Methods

2.1 Data Acquisition

Initially, 15 diffusion-weighted images (DWIs) were acquired using a 1.5T *Siemens Avanto* clinical scanner equipped with an eight-channel body coil. These placenta had an average gestational age $GA = (25.5 \pm 5.1)$ *weeks*. A diffusion-weighted echo-planar imaging spin echo sequence was employed with a repetition time (TR) of 3900 *ms* and

an echo time (TE) of 74.8 *ms*. The acquisition parameters included a bandwidth of 1184 *Hz/px*, a matrix size of 192×192 , and a field of view of 220×220 *mm*². Slice thickness was set at 5 *mm*, with 18 to 30 slices acquired depending on the size of the placenta.

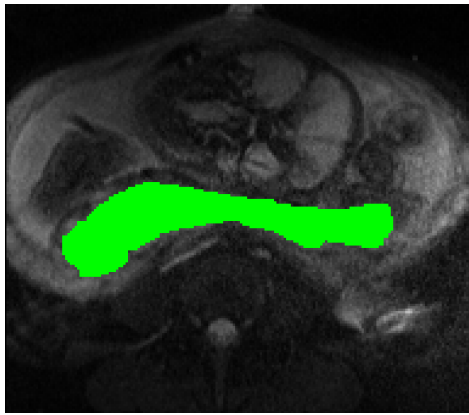
Then 7 DWIs for the other placentas, with an average gestational age of $GA = (26.3 \pm 4.0)$ *weeks*, were acquired using a 3*T* Siemens Prisma clinical scanner. A diffusion-weighted echo-planar imaging spin echo sequence was employed with a repetition time of 12300 *ms* and an echo time of 54 *ms*. The acquisition parameters included a bandwidth of 2437 *Hz/px*, a matrix size of 228×184 , and a field of view of 380×306 *mm*². Slice thickness was set at 3.5 *mm*.

Diffusion encoding gradients were applied along three non-co planar directions, using 10 different *b*-values ranging from 0 to 1000 *s/mm*² (increments: 0, 10, 30, 50, 75, 100, 200, 400, 700, and 1000 *s/mm*²). Signal averaging was performed over the three directions, with a total of 4 signal averages, $NS_{1.5T} = 4$ for the 1.5*T*, and $NS_{3T} = 2$ for the 3*T*, acquired for each *b*-value. Consequently, the entire imaging protocol lasted approximately 15 minutes [1].

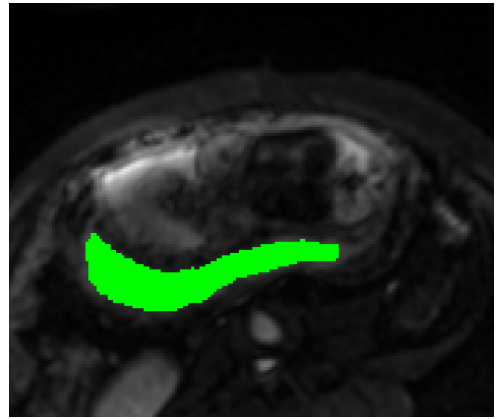
2.2 Pre-processing

2.2.1 Full placenta preliminary study

Once the DWI images have been taken, we can start with an accurate data analysis on the 15 placentas acquired at $B_0 = 1.5T$; the whole analysis process will be repeated for the 7 placentas acquired at $B_0 = 3T$.



(a) Segmentation of a placenta at 1.5*T*.



(b) Segmentation of a placenta at 3*T*.

Figure 6: Segmentation of the full placenta, for the two different values of magnetic field, taken at a *b*-value of $b = 200$ *s/mm*²

The first step is to segment one entire placenta, as shown in Figure 6a for the 1.5*T*

and 6b for the 3T. The segmentation process involves applying a mask to the DWI data to isolate the region of interest. The mask is used to filter out the relevant parts of the image, effectively zeroing out the areas outside the placenta.

The normalization is performed by multiplying the DWI data with the mask for each diffusion-weighted image, ensuring that only the relevant voxels within the placenta are considered. This step results in a dataset where the non-placental regions are set to zero, facilitating focused analysis on the placental tissue.

The next step involves calculating the mean and standard error of the signal within the masked ROI. This provides a quantitative measure of the signal within the placenta, which is crucial for subsequent analysis (refer to listing 1, in Appendix A).

Following the segmentation and normalization, the fitting process begins. The three models already discussed in section 2.3.1 are used to fit the data and extract the relevant parameters.

For each model, the fitting process is conducted “voxel per voxel”, across the entire placenta. Proper initial and boundary conditions are set for the parameters to ensure realistic and physiologically relevant estimates. The fitting algorithms are applied to each voxel, and parameters are extracted based on the best fit to the experimental data (see listing 3 as a code prototype for fitting).

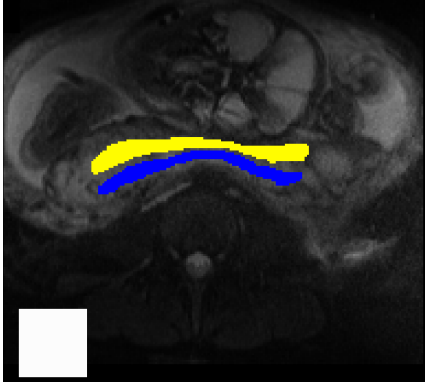
Moreover, maps are generated (through the function `plot_heatmap`, showed in list. 2) to visualize the spatial distribution of the fitted parameters across the placenta. These maps (which are displayed in section 3.4 through figures 12 - 13 - 14 for the 1.5T placentas and in figures 15 - 16 - 17 for the 3T) provide an intuitive and detailed view of the heterogeneity in the placental tissue, highlighting areas with different diffusion and perfusion characteristics.

2.2.2 Fetal and Maternal ROIs Extraction and Analysis

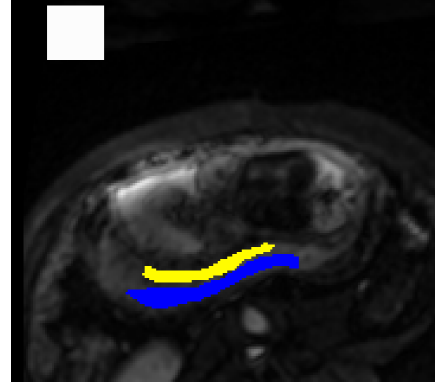
After segmenting the entire placenta, the next step involves isolating and analyzing different regions of interest (ROIs) within the placenta. Specifically, we focus on the fetal, maternal, and background regions by applying respective masks to the DWI data; refer to figure 7.

The DWI data is masked to isolate the fetal, maternal, and background regions. For each region, the mean signal and the standard error are calculated, providing a quantitative measure of the signal within each portion.

Next, the monoexponential model (eq. 7) is fitted to the mean DWI signal from the fetal and maternal ROIs to estimate the diffusion coefficient (D) and the signal at $b = 0$. The fitting involves setting initial parameters and bounds, performing the fit with the `curve_fit` imported from the `scipy.optimize` library, and calculating the mean squared error (MSE) and Akaike Information Criterion (AIC) to evaluate the fit quality. In a completely analogue way, the IVIM (eq. 8) and two-compartment



(a) Segmentation, with $B_0 = 1.5T$.



(b) Segmentation, with $B_0 = 3T$.

Figure 7: Placentas segmentation - with fetal ROI in yellow, maternal in blue, and background zone in white - for two placentas acquired at $b = 200 \text{ s/mm}^2$, for the two different values of B_0 .

perfusion models (eq. 6) are also applied, in order to find out the other relevant parameters.

2.3 Fitting controls

2.3.1 The Akaike information criterion

To mitigate the risk of overfitting due to the numerous parameters required for estimation in the two-perfusion IVIM model, precautions are necessary. In order to assess and compare the effectiveness of the three models, the Akaike Information Criterion (AIC) is employed.

AIC is a measure used in statistics to assess the goodness of fit of a statistical model. It balances the trade-off between the goodness of fit and the complexity of the model by penalizing models with more parameters. Lower AIC values indicate a better balance between fit and complexity, suggesting a more suitable model. It is expressed as [7]:

$$AIC = N_b \cdot \log(\text{MSE}) \cdot \frac{2k(k+1)}{N_b - k - 1} \quad (9)$$

Here, N_b denotes the count of b -values (so in our case it will be equal to 4 for the mono-exponential fit, since we are using the last 4 values of the 10 b -values supplied, while $N_b = 10$ for IVIM and two-perfusion), k represents the number of parameters in the model and MSE represents the mean squared error [7]:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The comparison of model goodness is facilitated by evaluating the discrepancy between the AIC values of the competing models (e.g., $AIC_{\text{IVIM}} - AIC_{\text{2perf}}$). A negative

outcome suggests that the former model (in this case, the two-perfusion model) is more adept at describing the data.

Furthermore, Akaike weights were computed for each model using the formula [7]:

$$w_i(AIC) = \frac{e^{-\frac{1}{2}\Delta_i(AIC)}}{\sum_{k=1}^K e^{-\frac{1}{2}\Delta_k(AIC)}} \quad (10)$$

where $\Delta_i(AIC) = AIC_i - \min(AIC)$.

2.3.2 The Kolmogorov-Smirnov test

In the field of data analysis, checking the goodness of fit is a crucial step in determining how well a mathematical model succeeds in describing experimental data. One of the most widely used statistical tools for this purpose is the chi-square test (χ^2). However, in cases where the sample size is small, such as in this study with only 10 b-values, the Kolmogorov-Smirnov (KS) test is often preferred. This preference is due to the KS test's ability to assess the goodness of fit without the need for binning data into categories, making it more suitable for continuous data and small sample sizes [8].

The Kolmogorov-Smirnov test is based on the maximum distance between the empirical distribution function of the observed data and the cumulative distribution function of the model. Mathematically, it is defined as [8]:

$$D = \sup_x |F_{\text{obs}}(x) - F_{\text{model}}(x)| \quad (11)$$

where D is the KS statistic, $F_{\text{obs}}(x)$ is the empirical distribution function of the observed data and $F_{\text{model}}(x)$ is the cumulative distribution function of the model.

A lower value of D indicates a better fit between the observed data and the model, while a higher value suggests a significant discrepancy. The KS test then provides a p -value $P(D)$, which represents the probability that the observed data could have been drawn from the model distribution. A p -value greater than 5% indicates that the model fits the data well, while a p -value less than 5% suggests that the model does not fit the data adequately.

We will find that $P(D) > 5\%$ for every model, concluding that each model actually adapts well to the data.

In this study, we have been able to perform the KS test thanks to the library imported in the *Google Colaboratory* environment: `from scipy.stats import kstest`.

3 Results and discussions

3.1 Signal-to-Noise Ratio

The Signal-to-Noise Ratio (SNR) is a crucial metric in medical imaging, as it compares the level of a desired signal to the level of background noise. A higher SNR indicates a clearer and more distinguishable signal, which is essential for accurate image analysis and interpretation. The SNR is particularly important in diffusion-weighted imaging (DWI), where the quality of the acquired signal can significantly impact the assessment of tissue properties. Mathematically, the SNR is defined as [2]:

$$\text{SNR} = \frac{\langle \text{ROI} \rangle}{\sigma(\text{background})} \quad (12)$$

where $\langle \text{ROI} \rangle$ represents the mean signal intensity within the region of interest and $\sigma(\text{background})$ represents the standard deviation of the noise in the background. This is a measure of the variability or fluctuation of the signal in an area that does not contain any useful signal, typically taken from regions outside the tissue of interest, see fig 7, white colored region.

To ensure a valid comparison of the signal-to-noise ratio (SNR) between the $1.5T$ and $3T$ acquisitions, we normalized the SNR values by taking into account the number of averages ($\text{NS}_{1.5T} = 4$ and $\text{NS}_{3T} = 2$) used in each acquisition. The SNR increases with the square root of the number of averages: $\text{SNR} \propto \sqrt{\text{NS}}$.

Thus, the normalized SNR values are calculated as [2]:

$$\text{SNR}_{\text{normalized}} = \frac{\text{SNR}_{\text{measured}}}{\sqrt{\text{NS}}} \quad (13)$$

We then calculated the **percentage gain** in SNR from $1.5T$ to $3T$ using the following equation:

$$\text{Gain} = \left(\frac{\text{SNR}_{3T}}{\text{SNR}_{1.5T}} - 1 \right) \cdot 100 \quad (14)$$

Figure 8 shows the percentage gain in normalized SNR for both fetal and maternal regions of interest (ROIs) across different b-values. The results indicate that the SNR gain is consistently higher for the $3T$ acquisitions compared to the $1.5T$ acquisitions. Specifically, the gain for the fetal ROI ranges from approximately 29% to 197%, while for the maternal ROI it ranges from approximately 81% to 201%. This significant increase in SNR at $3T$ suggests that higher magnetic field strength enhances the sensitivity and quality of the imaging, providing clearer and more detailed images of the placental structures.

In fig. 9, we show the behavior of the $\text{SNR}_{\text{normalized}}$ for the two ROIs taken into consideration, for each b -value used: fig. 9a for the $1.5T$ and fig. 9b for the $3T$ placentas.

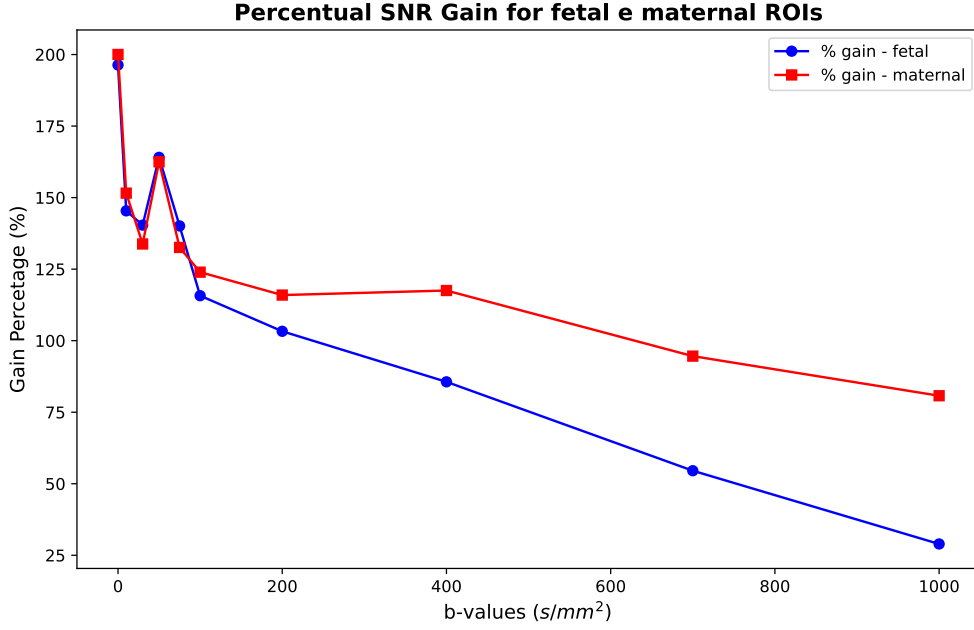


Figure 8: Percentage gain in normalized SNR for fetal and maternal ROIs across different b -values.

From the plots 9a and 9b, we can observe some key trends in the SNR measurements:

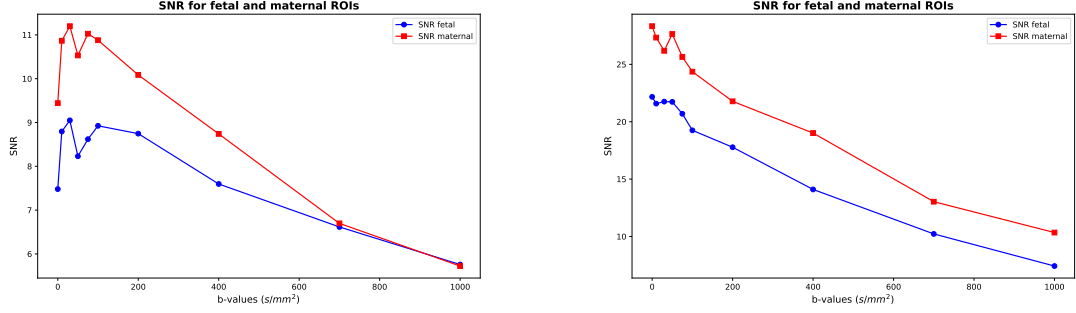
- **General trend:** The SNR generally increases with higher b -values for both fetal and maternal ROIs in both $1.5T$ and $3T$ environments. However, there are slight differences in the rate of increase and the maximum SNR values reached.

- **Differences between fetal and maternal normalized SNR:**

In both images, the SNR for maternal ROIs is consistently higher than that for fetal ROIs across all b -values. This indicates that maternal tissues provide a better signal compared to fetal tissues in MRI scans.

T_2 depends on B_0 , while diffusion in first approximation does not, and the fetal placenta is characterized by a shorter T_2 , leading to a faster decrease of the MR signal and therefore less time to collect high quality data.

- **Differences $1.5T$ and $3T$ normalized SNRs:** For the $3T$ acquisition, both fetal and maternal $SNR_{\text{normalized}}$ values are significantly higher compared to the $1.5T$ acquisition, nearly doubling in value due to the increase in magnetic field strength. This implies that the $3T$ acquisition provides superior SNR, due to the increased magnetic field strength (as the theory also suggests), which enhances the sensitivity of signal detection. Furthermore, the $SNR_{\text{normalized}}$ difference between fetal and maternal tissues is slightly more pronounced at $3T$, suggesting that the increased B_0 may also enhance the contrast between these tissue types.



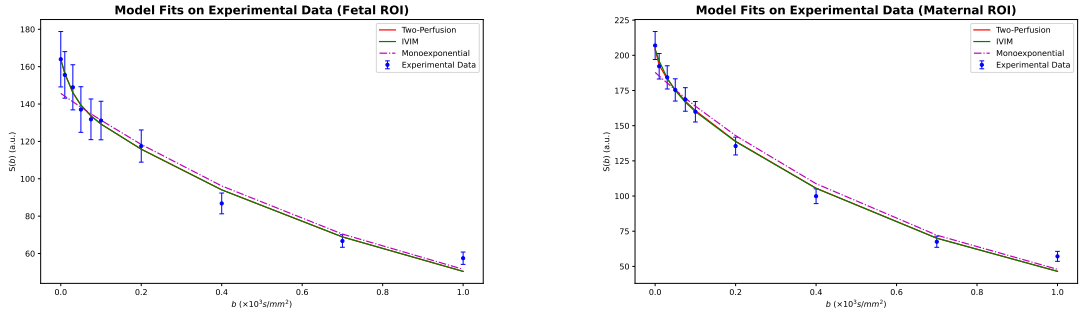
(a) Normalized SNR, fetal (blue) and maternal (red) - $1.5T$. (b) Normalized SNR, fetal (blue) and maternal (red) - $3T$.

Figure 9: Comparison between the two normalized SNR in fetal and maternal parts for $1.5T$ and $3T$ acquisitions.

3.2 Fitting the data

3.2.1 Fit for the $1.5T$ placentas

Akaike weights were computed for each model (the mono-exponential, IVIM, and two-perfusion). The higher-in value weight was the one associated to the IVIM model, for both the maternal and fetal ROIs. It was found to be $w_{\text{IVIM}}^{\text{fetal}}(AIC) = 0.80$ and $w_{\text{IVIM}}^{\text{maternal}}(AIC) = 0.83$, indicating a strong preference for this model over the others (refer to Fig. 10).



(a) Fit to DW-data obtained in the fetal ROI. (b) Fit to DW-data obtained in the maternal ROI.

Figure 10: Comparison of fits to DW-data in fetal and maternal ROIs.

The Akaike weights for the fetal (on the left, tab. 1) and maternal (on the right, tab. 2) ROI are shown below.

We can therefore conclude that the IVIM model is preferred over the other two.

Then, computing the KS test, we obtain the results stored in tables 3 and 4.

Being, in any case, $P(D) > 5\%$, we can conclude that our data fit well the three models, with a preference on the IVIM model, as previously found from the Akaike information criterion.

Model	AIC weights
$w_{2\text{perf}}(\text{AIC})$	0.19
$w_{\text{IVIM}}(\text{AIC})$	0.80
$w_{\text{mono}}(\text{AIC})$	3.1×10^{-5}

Table 1: Weights for the fetal ROI.

Model	AIC weights
$w_{2\text{perf}}(\text{AIC})$	0.17
$w_{\text{IVIM}}(\text{AIC})$	0.82
$w_{\text{mono}}(\text{AIC})$	3.9×10^{-3}

Table 2: Weights for the maternal ROI.

Model	D	$P(D)$
Two-Perfusion	0.22	0.60
IVIM	0.23	0.61
Mono-Exponential	0.27	0.39

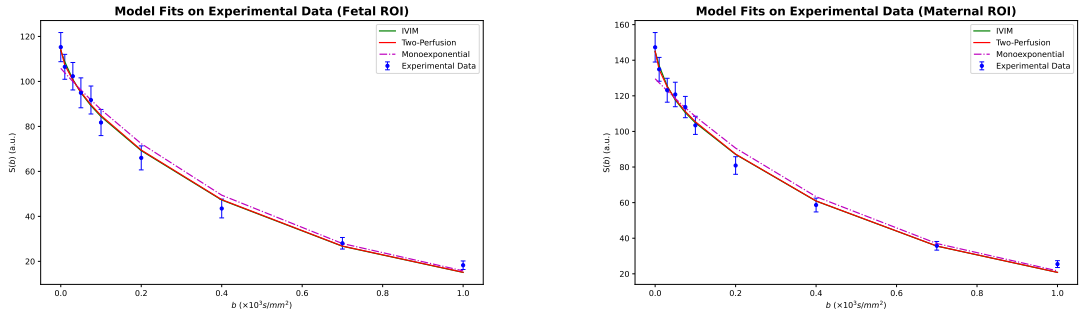
Table 3: KS test (fetal ROI, 1.5T).

Model	D	$P(D)$
Two-Perfusion	0.22	0.65
IVIM	0.22	0.66
Mono-Exponential	0.24	0.56

Table 4: KS test (maternal ROI, 1.5T).

3.2.2 Fit for the 3T placentas

Also in the case of the 3T placentas, the higher-in value weight was the one associated to the IVIM model, for both the maternal and fetal ROIs. It was indeed found to be $w_{\text{IVIM}}^{\text{fetal}}(\text{AIC}) = 0.87$ and $w_{\text{IVIM}}^{\text{maternal}}(\text{AIC}) = 0.82$, indicating again a strong preference for this model over the others. The fitting plots are shown in Fig. 11a and 11b.



(a) Fit to DW-data obtained in the fetal ROI. (b) Fit to DW-data obtained in the maternal ROI.

Figure 11: Comparison of fits to DW-data in fetal and maternal ROIs.

The Akaike weights for the fetal (on the left, Tab. 5) and maternal (on the right, Tab. 6) ROI are shown below.

We can therefore conclude that the IVIM model is preferred over the other two. Computing the KS test, we obtain the results stored in tab. 7 and 8.

Again, we always find $P(D) > 5\%$; we can therefore conclude that our data fit well the three models, with a preference on the IVIM model, as previously found from the Akaike information criterion.

Model	AIC weights
$w_{2\text{perf}}(\text{AIC})$	0.13
$w_{\text{IVIM}}(\text{AIC})$	0.86
$w_{\text{mono}}(\text{AIC})$	3.9×10^{-4}

Table 5: Weights for the fetal ROI.

Model	AIC weights
$w_{2\text{perf}}(\text{AIC})$	0.17
$w_{\text{IVIM}}(\text{AIC})$	0.82
$w_{\text{mono}}(\text{AIC})$	6.6×10^{-4}

Table 6: Weights for the maternal ROI.

Model	D	$P(D)$
<i>Fetal ROI</i>		
Two-Perfusion	0.19	0.77
IVIM	0.19	0.77
Mono-Exponential	0.20	0.73

Table 7: KS test (fetal ROI, $3T$).

Model	D	$P(D)$
<i>Maternal ROI</i>		
Two-Perfusion	0.18	0.80
IVIM	0.19	0.81
Mono-Exponential	0.21	0.70

Table 8: KS test (maternal ROI, $3T$).

3.3 1.5 T placentas parameters values

Below, through figures 12 - 13 - 14, we show the maps of the parameters obtained by fitting the three respective models, for the fifteen 1.5 T placentas segmented.

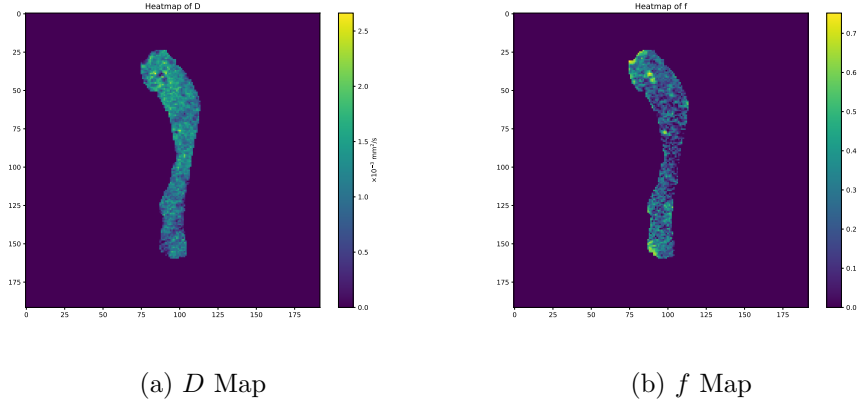


Figure 12: Maps for D and f .

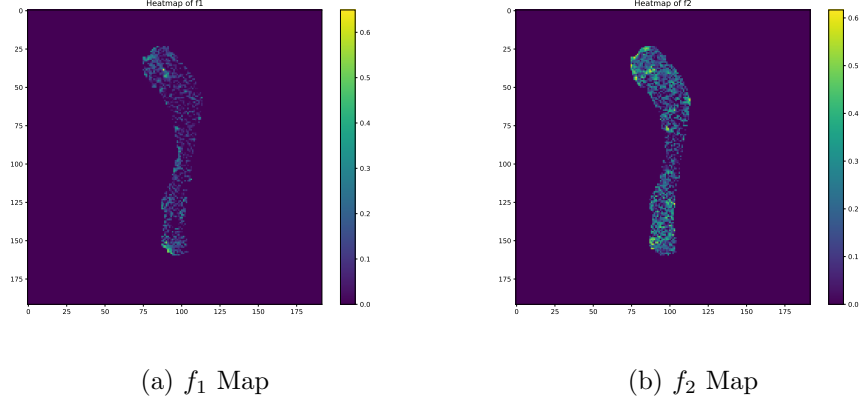


Figure 13: Maps for f_1 , and f_2 .

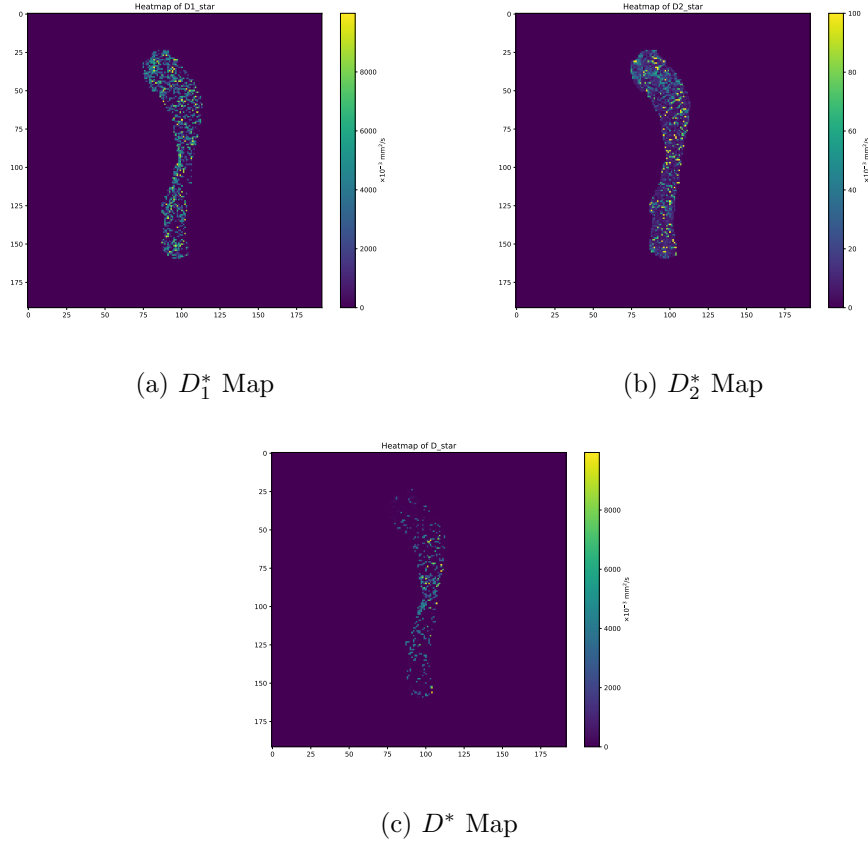


Figure 14: Maps for D^* , D_1^* and D_2^* .

As far as parameter f is concerned, from the map [12b](#), we can see that the fetal part of the placenta exhibits higher values of f , ranging from 0.4 to 0.7. In contrast, the maternal part predominantly shows lower f values, between 0.1 and 0.3. This gradient effectively delineates the fetal and maternal regions, suggesting a notable variation in tissue properties: in the fetal placenta there are more villi, so it is more sprayed and it is logical that it has a higher perfusion fraction f .

Moreover, the fetal part displays higher f_1 values, between 0.3 and 0.5, whereas

the maternal part shows lower values, generally around 0.1 to 0.3.

Likewise, concerning f_2 , the fetal part shows higher values, roughly between 0.3 and 0.5, while the maternal part has lower values, primarily from 0.1 to 0.3.

The observed gradients across these parameters collectively highlight the differences in tissue characteristics between the fetal and maternal parts of the placenta.

3.4 3 T placentas parameters values

Now, through figures 15 - 16 - 17, we show the maps of the parameters obtained by fitting the three respective models, for the other 7 ($3T$) placentas.

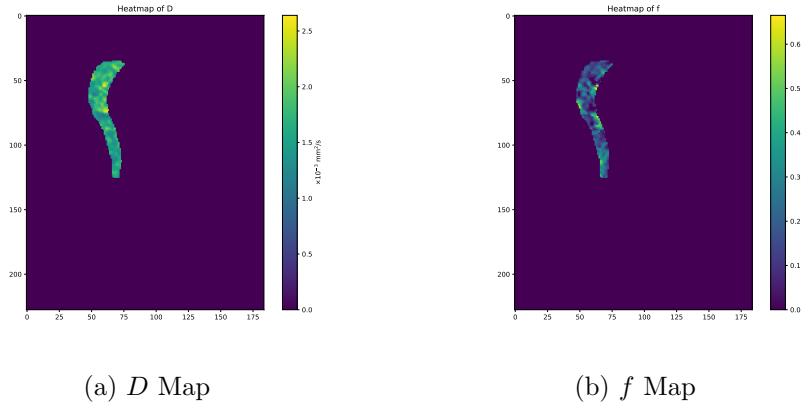


Figure 15: Maps for D and f .

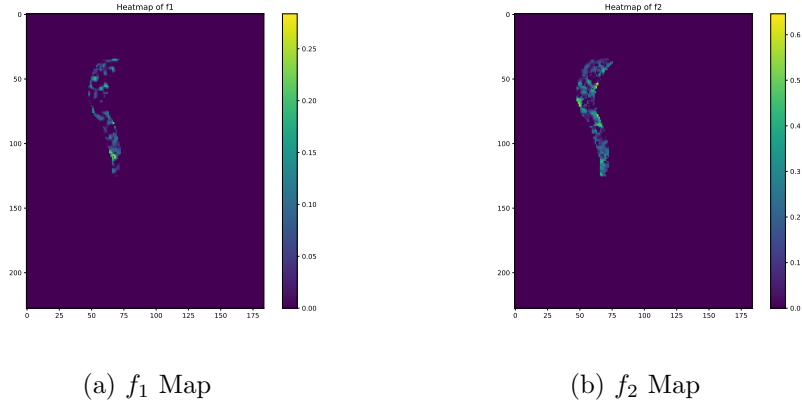


Figure 16: Maps for f_1 , and f_2 .

For parameter f , the fetal part of the placenta exhibited higher values, ranging from 0.4 to 0.6, while the maternal part predominantly showed lower values, between 0.0 and 0.3. This gradient effectively delineates the fetal and maternal regions, indicating differences in tissue properties between these parts.

Similarly, for parameter f_1 , the fetal part displayed higher values, approximately between 0.15 and 0.25, in contrast to the maternal part, which showed values generally

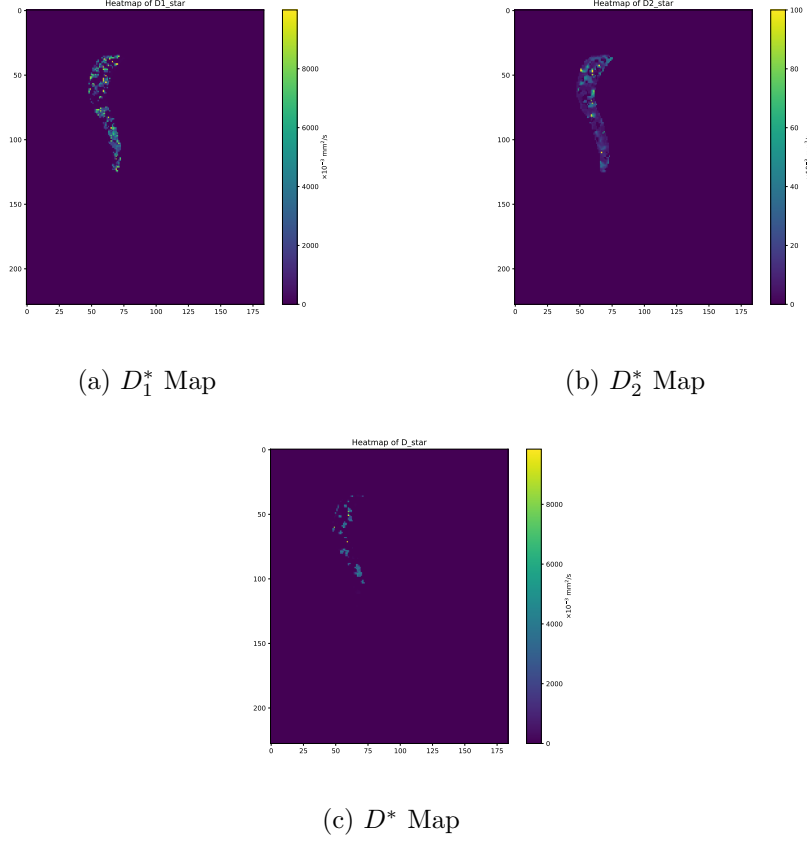


Figure 17: Maps for D^* , D_1^* and D_2^* .

around 0.05 to 0.15.

Additionally, the analysis of parameter f_2 revealed that the fetal part had higher values, roughly ranging from 0.3 to 0.5, whereas the maternal part had lower values, between 0.0 and 0.3.

The observed gradients across parameters f , f_1 , and f_2 collectively highlight the variations in tissue properties between the fetal and maternal parts of the placenta, underscoring the distinct physiological roles and characteristics of these regions.

3.5 Final parameters values

By fitting all the fifteen $1.5T$ placentas (and all the seven ones in the case of the $3T$), we obtain 15 (7) different values for each parameter, with associated errors.

By computing the average (and consequently propagating the error over the mean values) we obtain the results stored in tab. 9 for the $1.5T$ placentas and in tab. 10 for the $3T$ placentas.

We should point out the fact that the statistical errors in the measurements of f_1 and f_2 are quite substantial in both cases. This can be attributed to the complexity of the fitting process involved. Moreover, the chosen Regions of Interest (ROI) for

Parameter	<i>Maternal</i>	<i>Fetal</i>
	mean $\pm$ error	mean $\pm$ error
$D(\times 10^{-3} mm^2/s)$	1.574 ± 0.032	1.567 ± 0.041
f	0.163 ± 0.013	0.203 ± 0.016
f_1	0.020 ± 0.324	0.017 ± 0.215
f_2	0.144 ± 0.322	0.184 ± 0.216

Table 9: Parameters mean value for maternal and fetal ROI, in $1.5T$ acquisition.

Parameter	<i>Maternal</i>	<i>Fetal</i>
	mean $\pm$ error	mean $\pm$ error
$D(\times 10^{-3} mm^2/s)$	1.579 ± 0.049	1.552 ± 0.058
f	0.161 ± 0.019	0.187 ± 0.023
f_1	0.016 ± 0.661	0.018 ± 0.376
f_2	0.144 ± 0.670	0.169 ± 0.367

Table 10: Parameters mean value for maternal and fetal ROI, in $3T$ acquisition.

these measurements are quite extensive and lack homogeneity, further contributing to the observed variability in the parameter estimates. These factors collectively lead to significantly large errors in the parameter values obtained, impacting their precision.

As expected from theory, D does not change as the magnetic field strength changes f , f_1 and f_2 could be affected by T_2 , changing with the magnetic field.

Nevertheless, we have largely recovered the expected values for the parameters. For instance, as hypothesized at the beginning, parameter f is higher on the fetal side of the placenta, reflecting the increased metabolic activity and nutrient exchange necessary for fetal development. Similarly, the slowest perfusion parameter f_2 is higher on the fetal side, as expected, due to the active involvement of trophoblastic cells in nutrient exchange.

Conversely, on the maternal side, both f and f_2 values are lower, consistent with its primary role of supplying blood to the placenta rather than engaging directly in nutrient exchange. These results support our initial expectations and provide insight into the distinct physiological roles of the maternal and fetal regions of the placenta.

Regarding the diffusion coefficient D , the values are relatively consistent between the maternal and fetal parts, in both acquisitions. The slight differences observed suggest that diffusion is relatively uniform across the placenta, reflecting similar tissue density and water content in both regions.

3.5.1 Correlations between Parameters and Gestational Age

Gestational Age (GA) is a term used to describe the duration of pregnancy measured from the first day of the woman's last menstrual period to the current date. It is usually expressed in weeks. GA is crucial in obstetrics as it helps in assessing the growth and development of the fetus.

Monitoring the behavior of various physiological parameters with respect to GA can provide valuable insights into both maternal and fetal health. Analyzing the correlation between these parameters and GA using Pearson correlation can help in early detection of issues and guide clinical decisions.

The results of the Pearson correlation analysis are presented in tab. 11, while in figure 18 we show the trends of the 4 parameters f , f_1 , f_2 and D as functions of GA.

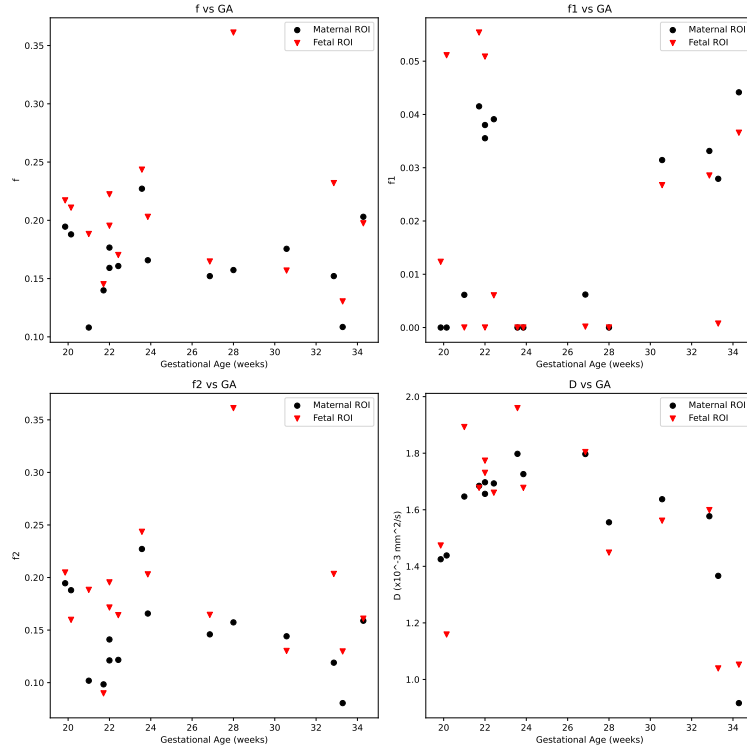


Figure 18: Trends of the 4 relevant parameters as functions of the GA (1.5T).

From the graphs concerning, the parameters f , f_1 and f_2 we do not observe a specific behavior: there seems to be no correlation.

On the other hand, parameter D seems to exhibit a **nearly quadratic trend** (similar to a negative parabola), particularly for the fetal ROI, indicating a decrease followed by an increase as GA progresses. This pattern suggests a non-linear relationship between D and GA, warranting further investigation into its potential clinical implications.

As far as Pearson correlation (also known as Pearson's r) is concerned, we know it measures the strength and direction of the linear relationship between two continuous variables, ranging from -1 to 1 . A low p -value, typically $P(r) < 0.05$, suggests that the correlation is statistically significant.

Parameter	<i>Maternal</i>		<i>Fetal</i>	
	r	$P(r)$	r	$P(r)$
f vs GA	-0.11	0.70	-0.04	0.89
f_1 vs GA	0.35	0.20	-0.05	0.86
f_2 vs GA	-0.25	0.37	-0.02	0.95
D vs GA	-0.49	0.07	-0.49	0.06

Table 11: Correlation values r and p -values ($1.5T$).

Notably, only D vs GA has a relatively stronger negative correlation for both maternal and fetal measurements, approaching statistical significance. Other parameters exhibit weaker correlations and higher p -values, indicating less significant linear relationships.

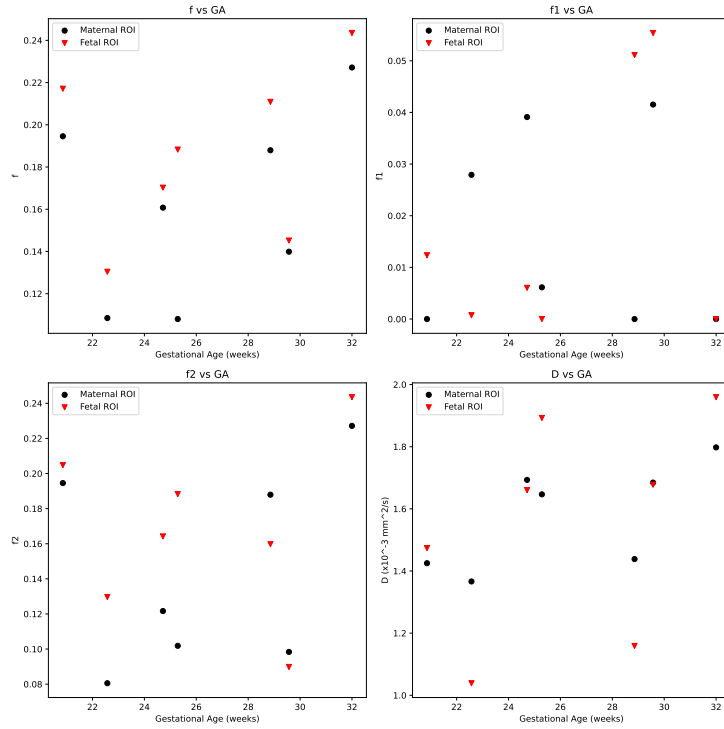


Figure 19: Trends of the 4 relevant parameters as functions of the GA ($3T$).

By conducting the same analysis on the $3T$ placentas, we find the graphs in fig. 19 and the results of the Pearson test showed in tab. 12.

Notably, the diffusion coefficient D vs GA shows a relatively stronger positive correlation for both maternal ($r = 0.65$) and fetal ($r = 0.42$) measurements, although the p -values indicate that these correlations are not statistically significant. The maternal D value approaches significance with $P(r) = 0.11$. Other parameters exhibit weaker correlations and higher p -values, indicating less significant linear relationships. These observations suggest that while there is some indication of a relationship between D and GA, the overall correlations for other parameters are less pronounced.

Parameter	<i>Maternal</i>		<i>Fetal</i>	
	<i>r</i>	<i>P(r)</i>	<i>r</i>	<i>P(r)</i>
<i>f</i> vs GA	0.41	0.36	0.33	0.48
<i>f</i> ₁ vs GA	−0.07	0.88	0.40	0.37
<i>f</i> ₂ vs GA	0.34	0.45	0.07	0.89
<i>D</i> vs GA	0.65	0.11	0.42	0.35

Table 12: Correlation values *r* and *p*-values (*3T*).

3.6 1.5*T* vs. 3*T*: Welch’s *t*-test analysis

Welch’s *t*-test is a statistical test used to compare the means of two independent samples. Unlike the Student’s *t*-test, Welch’s one does not assume that the two populations have equal variances, making it more robust in situations where this assumption is violated.

The goal of applying Welch’s *t*-test in this context is to statistically evaluate whether there are significant differences in the estimated parameters between the 1.5*T* and 3*T* segmented placentas.

Specifically, we aim to determine if the different magnetic field strengths have a significant impact on the measured values of *f*, *D*, *f*₁, and *f*₂. This analysis is crucial for understanding whether the magnetic field strength affects the diffusion and perfusion characteristics of the placenta, which can have implications for the interpretation of DWI data in clinical and research settings.

Mathematically speaking, the test statistics *t* is calculated as follows:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (15)$$

where $\bar{X}_1$ and $\bar{X}_2$ are the sample means, σ_1^2 and σ_2^2 are the sample variances and n_1 and n_2 are the sample sizes.

The degrees of freedom ν for the test are approximated using the following formula:

$$\nu \simeq \frac{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)^2}{\frac{(\sigma_1^2/n_1)^2}{n_1-1} + \frac{(\sigma_2^2/n_2)^2}{n_2-1}} \quad (16)$$

The calculated *t*-statistic is then compared against the critical value from the *t*-distribution with ν degrees of freedom to determine the *p*-value and assess the significance of the difference between the two sample means.

The results of the Welch’s t-test between the parameters obtained from $1.5T$ and $3T$ are shown in Table 13.

Parameter	t -statistic	p -value
Fetal Parameters		
D	8.29	2.26×10^{-16}
f	18.32	9.22×10^{-39}
f_1	-0.40	0.69
f_2	1.56	0.12
Maternal Parameters		
D	-2.60	0.01
f	3.92	9.34×10^{-5}
f_1	-0.17	0.86
f_2	-0.61	0.54

Table 13: Results of Welch’s t-test between the parameters obtained from $1.5T$ and $3T$.

For the **fetal parameters**, the diffusion coefficient (D) and perfusion fraction (f) exhibit highly significant differences between the two field strengths. These results suggest that the measurements for D and f are substantially different between $1.5T$ and $3T$ acquisitions. The cause is the greater field inhomogeneity at $3T$, which creates a bias in the estimation of D and f .

Conversely, the parameters f_1 and f_2 do not show significant differences, as indicated by p -values of 0.69 and 0.12, respectively (greater than 0.05 set as threshold).

For the **maternal parameters**, the diffusion coefficient (D) and perfusion fraction (f) also show significant differences between the two field strengths, with p -values of 0.01 and 9.34×10^{-5} , respectively. This indicates that the D and f parameters are significantly affected by the field strength in the maternal region. In contrast, the parameters f_1 and f_2 do not exhibit significant differences, with p -values of 0.86 and 0.54, respectively.

Overall, the Welch’s t -test results highlight that the field strength has a significant impact on certain parameters, particularly the diffusion coefficient (D) and perfusion fraction (f), for both fetal and maternal regions. However, the parameters f_1 and f_2 remain relatively unaffected by the change in field strength. This highlights the importance of considering field strength when interpreting DWI data in clinical and research contexts.

The differences in the diffusion coefficient (D) and the perfusion fraction (f) can be attributed to the intrinsic properties of the placental tissue and the way these properties are influenced by the magnetic field strength. The diffusion coefficient D measures the apparent diffusion of water molecules within the tissue. Higher field strengths provide greater sensitivity to subtle differences in the micro-structure and can better resolve small-scale variations in diffusion properties, leading to more precise and potentially higher D values.

Similarly, the perfusion fraction f represents the proportion of the signal attributable to micro-circulation within the tissue. Higher magnetic fields enhance the SNR, improving the accuracy of perfusion measurements. This improved SNR at higher field strengths allows for a more detailed assessment of the blood flow within the placental compartments, resulting in more accurate and potentially different f values compared to lower field strengths. Therefore, when analyzing DWI data, it is crucial to consider the field strength used in order to accurately interpret the diffusion and perfusion characteristics of the placenta.

4 Conclusions

In this study, we explored the diffusion and perfusion characteristics of both fetal and maternal regions of the placenta using diffusion-weighted imaging (DWI) with 1.5T and 3T MRI scanners. By employing monoexponential, IVIM, and two-compartment perfusion models, we derived various parameters to characterize these regions.

Our analysis identified significant differences in diffusion and perfusion properties between fetal and maternal tissues. The fetal region exhibited higher values for both the total perfusion fraction f and the slowest perfusion fraction f_2 . This suggests enhanced metabolic activity and nutrient exchange critical for fetal development. On the other hand, the maternal region showed lower values, aligning with its role in supplying blood to the placenta rather than directly engaging in nutrient exchange.

Additionally, our study demonstrated - as expected - that magnetic field strength significantly affects the signal-to-noise ratio. The 3T scans provided significantly higher SNR values compared to the 1.5T scans, nearly doubling due to the increased magnetic field strength. This improvement in SNR at 3T enhances signal detection sensitivity, improving the resolution and quality of the imaging data.

Using Welch's t -test, we found that the diffusion coefficient D and perfusion fraction f , exhibited relevant differences between the 1.5T and 3T acquisitions for both fetal and maternal tissues. We should investigate the reason of that, with further experiments,

since D and f - as they are defined - should not depend on the field B_0 . In particular as B_0 increases the field inhomogeneity increases and T_2 decreases.

Overall, our findings underscore the utility of high-field MRI in placental imaging. These insights into the distinct physiological roles and characteristics of fetal and maternal regions have potential implications for enhancing prenatal diagnostics and understanding placental function in both health and disease.

A Code Listings

```
1 # PART I: MASK APPLICATION
2 masked_dwi_data = np.zeros(dwi_data.shape)
3 dwi_norm = np.zeros(dwi_data.shape)
4
5 for i in range(10):
6     dwi_norm[:, :, :, i] = dwi_data[:, :, :, i] * mask_data
7
8 bool_mask = mask_data != 0
9 roi = dwi_data[bool_mask, :]
10 mean_roi = np.mean(roi, axis=0)
11 error_roi = np.std(roi, axis=0)/np.sqrt(roi.shape[1])
```

Listing 1: Mask creation code.

```
1 # MAPS CREATION
2
3 colormap = 'viridis' # trying to maximize contrast
4
5 def plot_heatmap(param_map, title, filename, unit=''):
6     plt.figure(figsize=(10, 8))
7     plt.imshow(param_map[:, :, param_map.shape[2] // 2],
8               cmap=colormap, interpolation='nearest')
9     cbar = plt.colorbar()
10    if unit: # Add units of measure if specified
11        cbar.set_label(unit) # Unit of measure on axis
12    plt.title(title)
13    plt.savefig(filename)
14    plt.close()
```

Listing 2: Heatmap creation code.

```

1 init_perfusion = [101,10,0,-np.inf] # initial conditions
2 b_perfusion = b_values # using all 10 b_values
3
4 # Loop over each voxel for fitting the parameters, 2-PERFUSION class:
5 for i in range(mask_data.shape[0]):
6     for j in range(mask_data.shape[1]):
7         for k in range(mask_data.shape[2]):
8             if mask_data[i,j,k] != 0:
9                 try:
10                     init_perfusion[-1] = dwi_norm[i,j,k,0] #set S(0)
11                     if param_maps_IVIM['f'][i, j, k] == 0.0:
12
13                         bounds_perfusion = ([100, 5, 0, 0],
14                                                [10000, 100, param_maps_IVIM['f'][i, j, k],
15                                                 np.inf])
16                         perfusion_model = Perfusion(
17                             param_maps_monoexponential['D'][i, j, k],
18                             param_maps_IVIM['f'][i, j, k])
19                         popt, _ = curve_fit(
20                             perfusion_model.function2, b_perfusion,
21                             dwi_norm[i,j,k,],
22                             init_perfusion, bounds=bounds_perfusion)
23                         param_maps_perfusion['D1_star'][i, j, k] = popt[0]
24                         param_maps_perfusion['D2_star'][i, j, k] = popt[1]
25                         param_maps_perfusion['f1'][i, j, k] = popt[2]
26                         param_maps_perfusion['f2'][i, j, k] =
27                         param_maps_IVIM['f'][i, j, k] - popt[2]
28
29                     except RuntimeError:
30                         continue
31                     except ValueError:
32                         continue

```

Listing 3: Two-compartment perfusion code for fitting. A similar procedure has been made for the other two models.

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